UNIT 1 · 9 hours

Mesh and Nodal Analysis



Circuit Fundamentals

Basic Electrical and Electronics Engineering (EEE103)
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Learning objectives

- State Kirchhoff's current and voltage laws
- Write mesh equations for a planar network with independent voltage sources
- Write nodal equations using a reference node
- Choose between mesh and nodal analysis for a given circuit
- Solve for branch currents, node voltages and power

Syllabus context

Basic nodal and mesh analysis excited by independent DC voltage sources, Power and Energy. Generation of sinusoidal voltage, frequency, average value, RMS value, form factor and peak factor, phasor representation of alternating quantities.

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Why systematic circuit analysis?

Goal: find unknown voltages, currents and power in a network of sources and resistors.

Ohm's law alone is not enough once a circuit has several loops and sources.

Two systematic methods: mesh analysis (based on KVL) and nodal analysis (based on KCL).

Scope here: networks excited by independent DC voltage sources.

Same ideas model mechanical, thermal and hydraulic systems by analogy.

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Kirchhoff's laws recap

KCL (current law)

The algebraic sum of currents at a node is zero.

$\Sigma I(\text{in}) = \Sigma I(\text{out})$

KVL (voltage law)

The algebraic sum of voltages around any closed loop is zero.

$\Sigma \text{EMF} = \Sigma \text{IR drops}$

Sign convention

Current entering a resistor makes that end positive. Cross a source from − to + and the voltage rises.

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Mesh analysis: procedure

1

Mark meshes

Assign a clockwise loop current to each window of the network.

2

Mark polarities

Draw + and − on every resistor consistent with the loop current.

3

Apply KVL

Sum of EMFs = sum of IR drops. A shared resistor carries (I1 − I2).

4

Rearrange

Write as simultaneous equations in the loop currents.

5

Solve

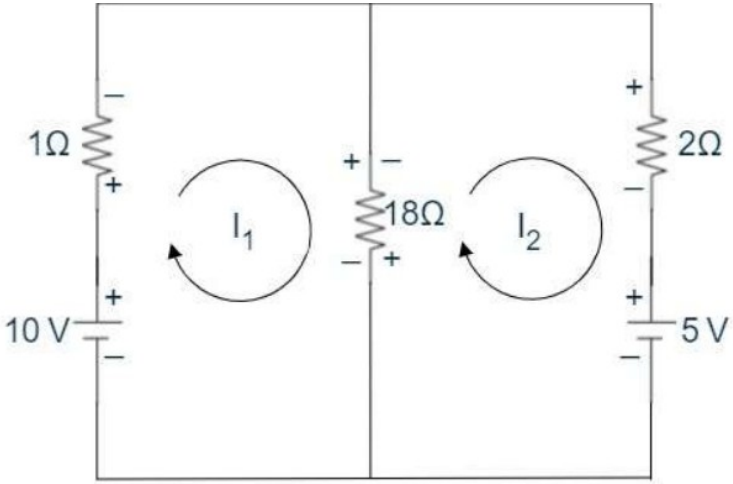
Branch current = algebraic sum of the loop currents through it.

A negative loop current only means the real current opposes the assumed direction.

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Mesh example: two-source network

- 10 V source with 1 Ω; shared 18 Ω; 2 Ω with 5 V source.
- Mesh 1: $10 - 1\cdot I1 - 18(I1 - I2) = 0$
- Mesh 2: $-18(I2 - I1) - 2\cdot I2 - 5 = 0$
- Rearranged: $-19\,I1 + 18\,I2 = -10$ and $18\,I1 - 20\,I2 = 5$
- $I1 = 1.964$ A, $I2 = 1.518$ A; $I(18\ \Omega) = 0.446$ A
- $P(18\ \Omega) = I^2R = 3.59$ W, $P(10$ V source$) = 19.64$ W



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Worked example: three-branch network

Problem 20 V and 30 V sources with 5 Ω, 15 Ω and a shared 8 Ω. Find the current in the 8 Ω resistor.

- Mesh 1: $20 = I1(5 + 8) - 8 \cdot I2 \rightarrow 13\,I1 - 8\,I2 = 20$
- Mesh 2: $-30 = I2(8 + 15) - 8 \cdot I1 \rightarrow -8\,I1 + 23\,I2 = -30$
- Determinant $\Delta = 13 \cdot 23 - 64 = 235$
- $I1 = (20 \cdot 23 - 240)/235 = 0.936$ A; $I2 = (13 \cdot (-30) + 160)/235 = -0.979$ A
- Current in 8 Ω = $I1 - I2$

Answer: 1.915 A in the 8 Ω resistor

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Nodal analysis: procedure

1

Choose reference

Pick a node as 0 V (ground), usually the one with most branches.

2

Label nodes

Give the other N − 1 nodes voltages V1, V2, …

3

Apply KCL

Write branch currents as (Va − Vb)/R and sum to zero at each node.

4

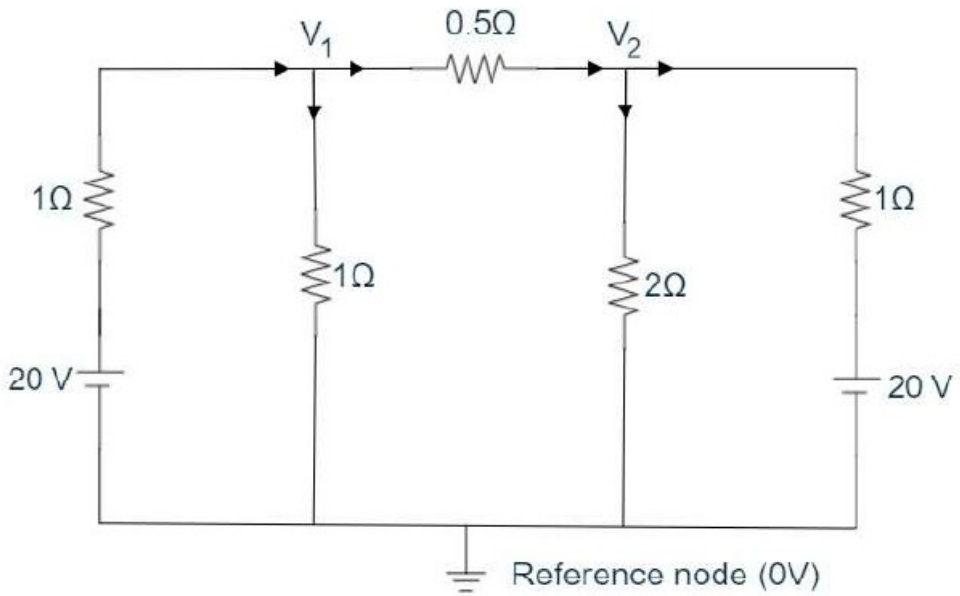
Solve

Solve N − 1 simultaneous equations for the node voltages.

Branch current = (voltage difference across the branch) / resistance.

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Nodal example



- Node 1: V1/1 + (V1 – V2)/0.5 = (20 – V1)/1
- → 4V1 – 2V2 = 20
- Node 2: (V1 – V2)/0.5 = (V2 – 20)/1 + V2/2
- → 2V1 – 3.5V2 = −20
- Solve: V1 = 11 V, V2 = 12 V

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Mesh versus nodal

Mesh analysis

- Uses KVL
- Unknowns: loop currents
- Equations = number of meshes
- Best for many series elements and voltage sources

Nodal analysis

- Uses KCL
- Unknowns: node voltages
- Equations = nodes − 1
- Best for many parallel branches and current sources

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Key takeaways

- KVL: sum of EMFs equals sum of voltage drops around a mesh
- KCL: currents at a node sum to zero
- Mesh analysis gives loop currents; branch current = algebraic sum of loop currents
- Nodal analysis gives node voltages with one node as reference
- Pick the method with fewer equations

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Practice questions

Q1. Using mesh analysis, find the current in each branch for 20 V and 30 V sources with 5 Ω, 8 Ω and 15 Ω resistors.

Q2. Find the node voltages V1 and V2 of the network in the nodal example.

Q3. A network has 5 nodes and 4 meshes. How many equations does each method need?

Q4. Why can a negative loop current appear in the solution?